MIL-M-51392A(EA)
14 November 1979
SUPERSEDING
MIL-M-51392(EA)
29 July 1974

MILITARY SPECIFICATION

METER, FLOW RATE INDICATING, FLOAT TYPE

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of flow rate indicating meter.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

Fed-Std-123 - Marking for Domestic Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810 - Environmental Test Methods.

FSC 6660

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL 5-15-4340 - Meter, Flow Rate Indicating, Float Type.
E136-91-1520 - Portable Gas Generator, Q219.
D136-44-160 - Tester, Air Flow and Leakage, Flowmeter, Air, Q195.

PUBLICATIONS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

136-305-1 - Procedure for Chemical Analysis of Chemical Agents Tabun (GA) and Sarin (GB).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 5-15-4340 and on the subsidiary drawings shall conform to the specifications listed thereon and to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the meter shall conform to the specifications and drawings listed on TDPL 5-15-4340 and subsidiary drawings.

3.2 Manufacture and assembly. The meter shall be manufactured and assembled as specified on Drawing D5-15-4340.

3.3 Leakage. With the air inlet blocked, leakage shall not exceed 0.60 cubic centimeter per minute when the interior air pressure is reduced to 16.0 ± 0.5 inch (40.6 ± 1.3 cm) of water below atmospheric pressure when tested as specified in 4.4.4.1.

3.4 Performance.

3.4.1 Airflow resistance. The pressure drop due to resistance to airflow through the air passages in the meter assembly shall be between 3.0 and 5.0 inches (7.6 and 12.7 cm) of water with an airflow of 1.10 ± 0.05 liter per minute when tested as specified in 4.4.4.2.

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3.4.2 Accuracy. When tested as specified in 4.4.4.3 at ambient conditions the meter accuracy shall be as follows:

(a) With the meter ball entirely within the zone outlined in green but tangent to the upper limit, the airflow through the meter shall be 1.50 (+0.10 or -0.10) liter per minute.

(b) With the meter ball entirely within the zone outlined in green but tangent to the lower limit, the airflow through the meter shall be 0.90 (+0.10 or -0.10) liter per minute.

3.5 Storage at low temperature. The meter shall meet the requirements of 3.3, 3.4.1, and 3.4.2 after storage at -60° ± 5°F (-51.1 ± 2.8°C) for at least 5 days and then at 75° ± 5°F (23.9 ± 2.8°C) for at least 3 hours when tested as specified in 4.3.3.1.

3.6 Storage at high temperature. The meter shall meet the requirements of 3.3, 3.4.1, and 3.4.2 after storage at 160° ± 5°F (71.1° ± 2.8°C) for at least 5 days and then at 75° ± 5°F (23.9° ± 2.8°C) for at least 3 hours when tested as specified in 4.3.3.2.

3.7 Operation at high relative humidity. When tested as specified in 4.3.3.3, after an exposure to a high relative humidity consisting of forcing air at a temperature of 70° ± 10°F (21.1° ± 5.6°C) and a relative humidity of 85 ± 5 percent through the meter at a rate of 1.2 ± 0.3 liter per minute for at least 12 hours but not more than 16 hours, the meter shall entrap chemical agent GA (see 6.3) from a gas stream containing chemical agent GA at a concentration of 10 ± 1 gamma per liter being forced through the meter at a rate of 1.0 ± 0.1 liter per minute for 1.0 ± 0.1 hour such that the quantity of chemical agent GA passing through the meter during the last 10.0 ± 0.1 minute of the test period does not exceed 0.40 gamma.

3.8 Operation after vibration. When tested as specified in 4.3.3.4, the meter shall meet the requirements of 3.3, 3.4.1, and 3.4.2 after exposure to the vibration environment described in MIL-STD-810, method 514, procedure VIII, curve Y (6,000 miles), applied along the longitudinal centerline of the meter and one axis perpendicular to the longitudinal centerline. There shall be no physical deterioration of the unit after vibration.

3.9 Preproduction. Prior to the start of regular production, a preproduction sample of meters shall be produced in accordance with this specification for examination and test (see 4.3).

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3.10 Workmanship. The meter shall be free from contamination (chips, dirt, oil or grease) and damage (dents, cracks, crazing, or deformations) which could impair it's use.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5, for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. Prior to the start of regular production, a preproduction sample of 12 meters shall be produced using the same materials, methods, equipments and processes as will be used in regular production.

4.3.2 Inspection procedures for examination and tests. The preproduction sample of meters shall be examined and tested in accordance with the classification of defects (4.4.3.2), and then tested in accordance with 4.3.3 and 4.4.4. Testing in accordance to 4.3.2 shall be accomplished as follows: Test the sample to 4.3.3.1 and 4.3.3.2, then divide the sample and test one half the sample to 4.3.3.4, then 4.3.3.3. The half tested to 4.3.3.3 is considered to be destructively tested.

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4.3.3 Testing. Tests shall be conducted as follows:

4.3.3.1 Storage at low temperature. Place the meter into a temperature test chamber. Both the meter and the test chamber must be at room temperature. Adjust the controls of the test chamber to produce the lower temperature required by 3.5. Maintain this temperature for at least 5 days. Then, adjust the controls of the test chamber to produce the higher temperature required by 3.5. Maintain this temperature for at least 3 hours. Remove the meter from the test chamber and subject it to the leakage, airflow resistance, and accuracy tests described in 4.4.4.1, 4.4.4.2, and 4.4.4.3. Leakage, airflow resistance, and accuracy shall be as required by 3.3, 3.4.1, and 3.4.2.

4.3.3.2 Storage at high temperature. Place the meter into a temperature test chamber. Both the meter and the test chamber must be at room temperature. Adjust the controls of the test chamber to produce the higher temperature required by 3.6. Maintain this temperature for at least 5 days. Then adjust the controls of the test chamber to produce the lower temperature required by 3.6. Maintain this temperature for at least 3 hours. Remove the meter from the test chamber and subject it to the leakage, airflow resistance, and accuracy tests described in 4.4.4.1, 4.4.4.2, and 4.4.4.3. Leakage, airflow resistance, and accuracy shall be as required by 3.3, 3.4.1, and 3.4.2.

4.3.3.3 Operation at high relative humidity. Mount the meter onto a support in a humidity test chamber such that the meter is vertical and its scale is visible from outside the test chamber. Connect the outlet of the meter, through a suitable throttling control, to a low pressure source. Adjust the controls of the test chamber to produce the temperature and relative humidity required by 3.7. Adjust the throttling control in the line between the meter and the low pressure source to produce the rate of flow through the meter required by 3.7. Permit the meter to remain undisturbed at the conditions of temperature, humidity, and flow just established for at least 12 hours but not more than 16 hours. Then, remove the meter from the humidity test chamber. Install the meter in series between a Q219 portable gas generator (E136-91-1520) producing the concentration of chemical agent GA required by 3.7 at the flow rate required by 3.7 and a suitable low pressure source. The meter must be installed so that the direction of flow through the meter during this test is the same as the direction of flow when the meter is being used to measure flow. Permit the flow through the meter to continue for 1.0 ± 0.1 hour. At 10.0 ± 0.1 minute before the end of this 1 hour period, install a suitable agent collector between the meter and the low pressure source. Analyze the contents of the agent collector in accordance with instruction manual 136-305-1. The agent collector shall not contain more than the quantity of chemical agent GA permitted by 3.7.

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4.3.3.4 Operation after vibration. Mount the meter onto a vibration exciter. The meter must not be supported by or attached to the mounting ixture except at the threaded plug (Drawing D5-15-4336). No pneumatic c nnections need be made; the meter need not be operated during the test. S bject the meter to the vibration environment required by 3.8. Then subject the meter to the leakage, airflow resistance, and accuracy tests described in 4.4.4.1, 4.4.4.2, and 4.4.4.3. Leakage, airflow resistance, and accuracy shall be as required by 3.3, 3.4.1, and 3.4.2. Examine the unit for physical deterioration.

4.3.4 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the meters offered for acceptance at one time which have been produced by one manufacturer at one plant under essentially the same manufacturing conditions and techniques and with no change in materials.

4.4.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedures.

4.4.3.1 For examination and nondestructive tests. Sample meters shall be examined and tested in accordance with the classification of defects 4.4.3.2 and MIL-STD-105.

4.4.3.2 Classification of defects.

(a) Meter, flow rate indicating, float type (Dwg D5-15-4340).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Leakage	4.4.4.1
102	Airflow resistance	4.4.4.2
103	Accuracy	4.4.4.3

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4.4.3.2 Classification of defects (continued).

AQL 2.5 percent defective

104	Component missing, incorrect, or incorrectly located
105	Calibration markings missing
106	Air passages blocked, missing, or mislocated
107	Workmanship (see 3.10)

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Leakage. Determine the leakage of the meter in accordance with the Q195 (D136-44-160) Air Flow and Leakage Tester.

4.4.4.2 Airflow resistance. Determine the airflow resistance of the meter in accordance with the Q195 Air Flow and Leakage Tester.

4.4.4.3 Accuracy. Determine the accuracy of the meter in accordance with the Q195 Air Flow and Leakage Tester.

5. PACKAGING

5.1 Packaging, packing and marking, interplant shipment (see 6.4). The meter shall be unit packed and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in conformance with Fed-Std-123 and shall include the contractors name, address and contract number.

5.2 Repair parts. When this item is being procured for storage and issue as a repair part, preservation, unit packing, packing and marking shall be as specified on the repair parts packaging data sheet which is identified by the National stock number.

6. NOTES

6.1 Intended use. The meter is used as a component of the detector unit which is used in a series of chemical agent alarms.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

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(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Chemical agent GA. As specified in the contract, chemical agent GA will be furnished to the contractor by the Government. Special facilities and safety precautions are required for the storage and handling of chemical agent GA.

6.4 Interplant shipment. Packaging and packing for interplant shipment is for supplies and material that do not enter the military supply system. Typical interplant shipments are shipments from a vendor to a contractor, or from a contractor to a military arsenal or plant.

6.5 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 251. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

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